
5. Exercise on QFT for many-body systems

Sommersemester 2026

TUTORIUM: Friday, 12.06.2026.

10. Van Hove singularities

$2+2+1+2^*=5+2^*$ points

Consider the dispersion relation (single-particle energy states) for electrons on a simple hyper-cubic lattice in d dimensions, with only nearest-neighbor hopping:

$$\varepsilon_{\mathbf{k}} = -2t \sum_{i=1}^d \cos k_i, \quad (1)$$

with the hopping amplitude t and the lattice constant $a = 1$. The density of single-particle states in this system is then given by

$$N(\epsilon) = \frac{1}{(2\pi)^d} \int_{-\pi}^{\pi} d^d k \, \delta(\epsilon - \varepsilon_{\mathbf{k}}). \quad (2)$$

In the first exercise you have calculated numerically and then plotted these densities of states for $d = 1, 2, 3$. Here, the singular structures (divergences, cusps) of these functions should be analyzed analytically.

- a) Calculate $N(\epsilon)$ for $d = 1$ explicitly and determine the interval $[\epsilon_1, \epsilon_2]$ on which $N(\epsilon) \neq 0$. Moreover, identify the values ϵ^* where D diverges, i.e. where $N(\epsilon^*) = \infty$. From which points $\mathbf{k}^*$ in the dispersion relation originate these divergences? Show that the divergences can be reproduced by taking into account only the contributions from these $\mathbf{k}^*$ -points. (*Hint: Replace $\varepsilon_{\mathbf{k}}$ in Eq. (2) by a corresponding Taylor-expansion around these points up to second order.*)
- b) For $d = 2$ one can show that $N(\epsilon)$ is essentially given by a complete elliptic integral of the first kind. Here, however, only the singular contributions to $N(\epsilon)$ should be analyzed. As in the one-dimensional case a singular contribution originates from stationary points in the dispersion relation. Determine the kind of stationary point (i.e., maximum, minimum or saddle point) which generates this so-called Van Hove singularity in the two-dimensional DOS and determine the singular contribution to $N(\epsilon)$ by expanding $\varepsilon_{\mathbf{k}}$ around corresponding stationary point in Eq. (2) as for the one-dimensional case in a).
- c) Try to predict how the singular behavior of the DOS evolves with the dimensions of the system for $d \geq 3$.
- d) (Bonus points) Finally, consider the limit $d \rightarrow \infty$. In this case, one has to rescale the hopping amplitude as $t \rightarrow \frac{t}{\sqrt{d}}$, in order to render the total energy of the system as well as the second moment (standard deviation) of the density of state finite. Show that $N_{\infty}(\epsilon)$ is proportional to a Gaußdistribution.

11. Magnetic susceptibilities in d dimensions

1.5+1.5+1+1=5 points

Consider a system of non-interacting electrons on a (hyper)cubic lattice whose energy dispersion is given by Eq. (1).

- a) Compute the *magnetic* susceptibility, i.e. the Fourier transform of the spin-spin response function $\langle T_\tau S_z(\mathbf{r}_i, \tau) S_z(0, 0) \rangle$, with $S_z(\mathbf{r}_i, \tau) = \frac{\hbar}{2} [c_{i\uparrow}^\dagger(\tau) c_{i\uparrow}(\tau) - c_{i\downarrow}^\dagger(\tau) c_{i\downarrow}(\tau)]$, for the frequency $\Omega_m = 0$ (static susceptibility), and for the two momenta $\mathbf{Q} = (0, 0, 0, \dots)$ (ferromagnetic susceptibility) and $\mathbf{Q} = (\pi, \pi, \pi, \dots)$ (antiferromagnetic susceptibility).
- b) Determine the leading divergences of the ferromagnetic and the antiferromagnetic susceptibilities for $T \rightarrow 0$ in $d=2$ dimensions. To this end write the total density of states as a sum of a singular and a regular contribution as calculated in 10 b).

*Hint: It is convenient to write the total density of states as: $\mathcal{N}(\varepsilon) = \mathcal{N}_{\text{reg}}(\varepsilon) + \mathcal{N}_{\text{sing}}(\varepsilon)$, and treat the two contributions separately. For the **ferromagnetic** case, expand $\mathcal{N}_{\text{reg}}$ around zero (Sommerfeld-like expansion) and, for the singular part, change variable to $x = \beta\varepsilon$ to make the temperature dependence manifest. For the **antiferromagnetic** case, (i) differentiate the susceptibility with respect to β before integrating, (ii) proceed as in the ferromagnetic case, (iii) integrate back over β at the end.*

- c) Discuss how the results of b) are modified in $d \geq 3$ dimensions.
- d) Consider now non-interacting electrons on a one-dimensional lattice with dispersion $\epsilon_k = -2t \cos(ka)$ at half-filling ($\mu = 0$). Is there a Q -point in the Brillouin zone, $Q \in [0, 2\pi]$, for which $\epsilon_{k+Q} = -\epsilon_k = 0$? What is the signature of this “nesting” property in the free (bubble) susceptibility $\chi_0(Q, \omega = 0)$ (calculated in Exercise 2, Problem 4c)? Remember that the sum over k , $\sum_k$, can be replaced by $\int d\epsilon \mathcal{N}(\epsilon)$ with the density of states $\mathcal{N}(\epsilon)$ from Exercise 1.